Jared Siegel

Princeton University, Department of Astrophysical Sciences — Princeton, NJ 08544

siegeljc@princeton.edu jaredcsiegel.github.io 0000-0002-9337-0902

Education

Princeton University — Expected 2027

PhD in Astrophysics | Thesis Advisors: Alexandra Amon and Jenny Greene

Research: Observational cosmology and galaxy evolution; weak lensing; baryonic feedback; intrinsic alignment of galaxies

University of Chicago — 2022

BA in Physics & BS in Astrophysics | Thesis Advisor: Leslie Rogers

Awards and Grants

Harold W. Dodds Fellowship | Princeton University — 2026–2027

"... recognizes exceptional students in their later years of study ... provides full university fellowship tuition and a 12-month premium stipend."

Graduate Research Fellowship | Jane Street — 2026–2027

"... supports exceptional doctoral students currently pursuing a PhD in electrical engineering and computer science, mathematics, physics, or statistics."

NSF Graduate Research Fellowship | National Science Foundation — 2022–2026

Centennial Fellowship | Princeton University — 2022–2027

Chambliss Astronomy Student Award | American Astronomical Society — 2020

Observing Proposals

PI James Webb Space Telescope Cycle 4 GO #8607 — 2025

"Studying cosmic noon at 200 parsec scales: resolved spectroscopy of a magnified dusty quiescent galaxy"

Peer Reviewed Publications

Summary: 13 first author; 2 major contributor; 21 total

Cosmology and Galaxies .....

8. **Siegel, J.**, et al., "Evidence for a thermal pressure deficit in galaxy groups from the tSZ effect and weak lensing", submitted to ApJ, arXiv:2608.00164

7. Bigwood L., McCullough J., **Siegel, J.**, et al., "Confronting cosmic shear astrophysical uncertainties: DES Year 3 revisited", submitted to PRD, arXiv:2512.04209

6. Bigwood L., Yamamoto M., **Siegel, J.**, et al., "The kinetic Sunyaev Zeldovich effect as a benchmark for AGN feedback models in hydrodynamical simulations: insights from DESI + ACT", submitted to MNRAS, arXiv:2510.15822

5. **Siegel, J.**, et al., "The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy lensing", MNRAS, 549, 4 (2026), arXiv:2512.02954

4. **Siegel, J.**, Amon A., et al., "Joint X-ray, kinetic Sunyaev-Zeldovich, and weak lensing measurements: toward a consensus picture of efficient gas expulsion from groups and clusters", The Astrophysical Journal, 1003, 2 (2026), arXiv:2509.10455

3. **Siegel, J.**, McCullough, J., Amon A., et al., "Intrinsic alignment demographics for next-generation lensing: Revealing galaxy property trends with DESI Y1 direct measurements", submitted to MNRAS, arXiv:2507.11530

2. **Siegel, J.** & Melchior, P., "Spatially Resolved Galaxy-Dust Modeling with Coupled Data-Driven Priors", The Astrophysical Journal, 986, 212 (2025), arXiv:2411.08111

1. **Siegel, J.**, Setton, D., Greene, J., et al., "UNCOVER: Significant Reddening in Cosmic Noon Quiescent Galaxies", The Astrophysical Journal, 985, 125 (2025), arXiv:2409.11457

Extrasolar Planets .....

5. **Siegel, J.**, Halverson, S., Luhn, J., Zhao, L., Al Moulla, K., et al., "Quiet Please: Detrending Radial Velocity Variations from Stellar Activity with a Physically Motivated Spot Model", The Astronomical Journal, 168, 158 (2024), arXiv:2408.07121

4. **Siegel, J.**, Winn, J., Albrecht, S., "Ponderings on the Possible Preponderance of Perpendicular Planets", The Astrophysical Journal Letters, 950, 1 (2023), arXiv:2305.13400

3. **Siegel, J.** & Rogers, L., "Mass Upper Bounds for Over 50 Kepler Planets Using Low-S/N Transit Timing Variations", The Astronomical Journal, 164, 139 (2022), arXiv:2208.14398

2. **Siegel, J.**, Rubenzahl, R., Halverson, S., & Howard, A., "Into the Depths: a new activity metric for high-precision radial velocity measurements based on line depth variations", The Astronomical Journal, 163, 260 (2022), arXiv:2204.05810

1. **Siegel J.** & Fabrycky, D., *Resonant Chains of Exoplanets: Libration Centers for Laplace Angles*, The Astronomical Journal, 161, 290 (2021), arXiv:2104.14665

High Energy Astrophysics .....

3. **Siegel, J.**, Kiato, I., Kalogera, V., Berry, C., Maccarone, T., et al., "Investigating the Lower Mass Gap with Low Mass X-ray Binary Population Synthesis", The Astrophysical Journal, 952, 212 (2023), arXiv:2209.06844

2. **Siegel, J.**, Dwarkadas, V. V., Frank, K. A., & Burrows, D. N., "Can the Fe K-alpha line reliably predict supernova remnant progenitors?", The Astrophysical Journal, 922, 67 (2021), arXiv:2109.01157

1. **Siegel, J.**, Dwarkadas, V. V., Frank, K., & Burrows, D. N., "Analysis of XMM-Newton Observations of Supernova Remnant W49B and Clues to the Progenitor", The Astrophysical Journal, 904, 175 (2020), arXiv:2010.04765

Students Supervised

Kate Shavelle | Princeton Graduate Student — Fall 2025—Present

Daryna Yushchenko | Princeton Undergraduate Student — Summer 2026—Present

Selected Talks

Seminar [*Invited*] — Harvard University, Apr. 2026
CMB/LSS Seminar — Cambridge University, Apr. 2026
Physics Seminar [*Invited*] — University of Alabama, Apr. 2026
Cosmology Seminar [*Invited*] — UC Berkeley, Mar. 2026
AGN & Energy Flows Workshop — Center For Computational Astrophysics, Feb. 2026
Mock NYC Workshop — Center For Computational Astrophysics, Jan. 2026
Cosmology Seminar [*Invited*] — Yale University, Nov. 2025
European Southern Observatory MMC2025 — Italy, Oct. 2025
Cosmic Ecosystems Conference — Perimeter Institute, Canada, Jul. 2025
Millimeter Universe Conference — University of Chicago, Jun. 2025
Big Data, Big Questions: The Future of Cosmological Surveys — Germany, May 2025

Outreach and Service

Princeton Prison Teaching Initiative — Spring 2025—Present

Developed curricula, served as an in-class tutor, and oversaw the teaching team for AST 207 "Gateway to Science" at East Jersey State Prison.

Teaching

Teaching assistant—Princeton University

AST 401 Cosmology — Spring 2026
APC 524 Software Engineering for Scientific Computing — Fall 2025
AST 207 Gateway to Science: Observational Astronomy — Spring 2025 AST 205
Planets in the Universe — Fall 2023

Teaching assistant—University of Chicago

ASTR 211 Computational Techniques in Astrophysics — Spring 2021
ASTR 205 Introduction to Python Programming — Fall 2020